

Talking about the hamsters

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March 6th, 2026

We discussed the following question [in the MCBDD course \(link to the course's website\)](#).

I have two pills and two hamsters. The pills are identical.

The two hamsters are optically identical, too, so that I cannot tell the difference.

I know that both pills are one of the following three types:

- * Pill A makes both hamsters sleep.
- * Pill B makes neither animal sleep.
- * Pill C makes one animal sleep but not the other.

Now I pick a pill, feed it to one hamster, and the hamster falls asleep.

What's the probability that the second pill makes the other animal sleep, too?

I think the answer is $2/3$. Below are two different ways to reach the conclusion.

A less formal method

It is clear that pill B is not possible, so the randomly chosen hamster must have been given either Pill A or Pill C.

The most tricky part of the game is that pill C makes one hamster sleep but not the other, and there is no way we can tell the difference between the two until we give them Pill C. For convenience, let us call the hamster that falls asleep upon treatment with Pill C 'H1' and the one which would have stayed awake 'H2'.

Our observation was that upon being given a pill, a hamster - either H1 or H2 - falls asleep. What are the possible scenarios?

- Pill A was given to H1
- Pill A was given to H2
- Pill C was given to H1

All three scenarios have an equal probability, i.e. $1/3$. Note that none of the events involve

Pill B, and the event that Pill C was given to H2 is impossible, because it contradicts the observation that the first hamster falls asleep.

- If Pill A was given to H1 (probability $1/3$), the probability that H2 (the other hamster) falls asleep when another pill is given is 1.
- If Pill A was given to H2 (probability $1/3$), the probability that H1 falls asleep when another pill is given is 1.
- If Pill C was given to H1 (probability $1/3$), the probability that H2 falls asleep when another pill is given is 0.

Therefore, the total likelihood that the other hamster falls asleep when another pill of the same type is given is $2/3$.

Intuitively, since Pill A always works and Pill C works only half the time, the observation that a hamster falls asleep is more likely to have been caused by Pill A than by Pill C.

The Bayes' theorem solution

Assigning probabilities to pill selection

Since we have no other information than the two pills are of the same type, we believe that the probabilities that the two pills are of Type A, B, or C are equal: $p(A) = p(B) = p(C) = \frac{1}{3}$.

Probability of the observation, i.e. a randomly chosen hamster falling asleep

Next, we can list how likely it is that a randomly chosen hamster falls asleep given each pill:

- Since Pill A makes both hamsters sleep, the probability that it makes a randomly chosen hamster sleep is 1.
- It follows that the probability of Pill B making a randomly chosen hamster sleep is 0.
- For Pill C, the probability that it makes a randomly chosen hamster sleep is $1/2$.

Therefore, if we call the event of a hamster falling asleep after taking a pill S , the total probability $p(S)$ is

$$p(S) = p(A)p(S|A) + p(B)p(S|B) + p(C)p(S|C) = 1/3 * 1 + 1/3 * 0 + 1/3 * 1/2 = 1/2$$

Finding the probability of having picked each pill given the observation

Now we can use Bayes' theorem to ask the reverse question: given that a randomly chosen hamster has fallen asleep, what are the probabilities that the pill is of type A, B, or C respectively?

$$p(A|S) = \frac{p(A)p(S|A)}{p(S)} = \frac{1/3 \cdot 1}{1/2} = 2/3$$

$$p(C|S) = \frac{p(C)p(S|C)}{p(S)} = 1/3 * (1/2)/(1/2) = 1/3$$

$$p(B|S) = 0$$

Namely, given that we observe a randomly chosen hamster falling asleep, we believe that the probability that we have given a pill of type A is 2/3, and that of type C is 1/3.

Finding the probability that the other hamster sleeps with another pill of the same type

Let us denote the event that the other hamster sleeps, too, as T .

Note that there are only two possibilities of T happening, either by giving Pill A or by giving Pill C. In other words, $p(T|S) = p(T|A)p(A|S) + p(T|C)p(C|S)$.

- Since Pill A makes both hamsters sleep, $p(T|A) = 1$.
- Since Pill C makes one sleep but not the other, $p(T|C) = 0$.

Therefore,

$$p(T|S) = 2/3 * 1 + 1/3 * 0 = 2/3$$

Namely, the probability that the other hamster, given another pill of the same type, would fall asleep is 2/3.

Some more thoughts and relevant information

The game was a variant of [the Monty Hall problem](#).

Imagine that Pill C has a probability of 1/100,000 of making a hamster fall asleep. Does this make the result more intuitive? That is, it is far more likely that Pill A was picked at random, and therefore the other hamster is more likely to fall asleep upon the treatment with the same Pill.

You are welcome to pose the question to large language models such as ChatGPT, Claude, and DeepSeek. I observed some quite interesting patterns.

If you still have questions or concerns, please feel free to reach out by email or in person at any time.